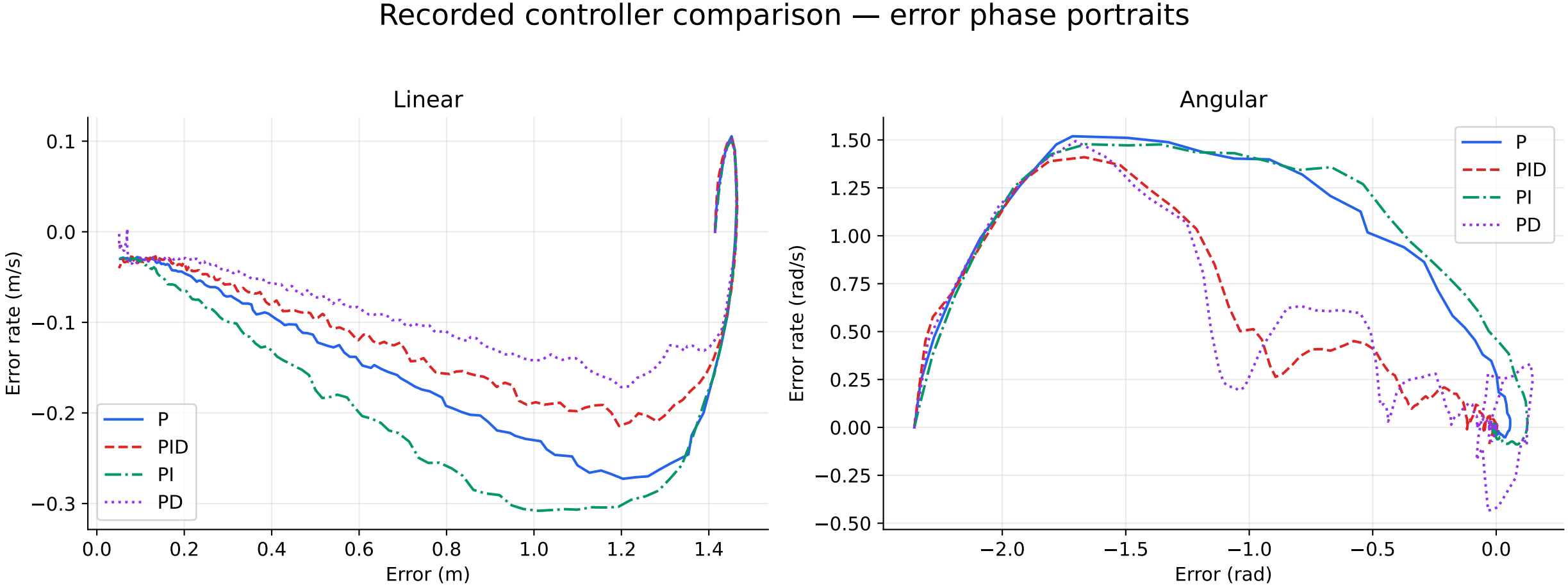
Recorded controller comparison — error phase portraits
Linear
Angular
Error rate (m/s)
Error (m)
Error rate (rad/s)
Error (rad)
P
PID
PI
PD
0.1
0.0
−0.1
−0.2
−0.3
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−2.0
−1.5
−1.0
−0.5
0.0